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AERODYNAMIC MODELING OF A DUAL-SAIL SPHERICAL SAILING OMNIDIRECTIONAL ROVER (SSAILOR)

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ABSTRACT

Conventional sailboats achieve high aerodynamic efficiency by pairing a main sail with a jib sail, forming a two-sail system that creates a slot effect, accelerates airflow, and significantly enhances lift. The Spherical Sailing Omnidirectional Rover (SSailOR) adapts this same aerodynamic principle to a radically different platform: a lightweight, rolling spherical rover designed for navigation in remote and harsh environments. For SSailOR, the dual sail arrangement is essential not only for generating forward thrust but also for providing heading control and the ability to maneuver relative to the prevailing wind direction. Understanding how the two sails interact aerodynamically is therefore critical for determining achievable lift, overall efficiency, and control authority of the rover.

This study investigates the aerodynamic behavior of SSailOR's main-jib sail configuration using a numerical implementation of the Weissinger method to compute aerodynamic forces for a combination of sails. Further, a verification study was performed with XFLR5. The goal is to develop and validate a computational framework capable of capturing the coupled aerodynamic effects generated by the two sails when operating together. The analysis focuses on understanding how sail planform, sail trim, spacing, and angle combination influence lift amplification, drag penalties, and strength of the slot effect.

The Weissinger model successfully captures these trends and shows strong agreement with XFLR5 across a broad set of operating conditions. The validated model offers a fast and reliable computational tool for exploring geometric variations, optimizing sail area and placement, and guiding future SSailOR design decisions. Overall, this work provides a verified aerodynamic modelling framework for SSailOR's dual sail system and contributes to a broader understanding of coupled sail aerodynamics in both conventional sailing applications and emerging wind-propelled robotic platforms.

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Keywords: Dual-sail aerodynamics, low-fidelity CFD modeling, aerodynamic force prediction, lift-drag optimization, numerical modeling, XFLR5, vortex lattice method, SSailOR

NOMENCLATURE

Roman letters

C_L	Coefficient of Lift
C_{Di}	Coefficient of Induced Drag
C_p	Coefficient of Pressure
D	Aerodynamic drag force [N]
F	Aerodynamic Influence Coefficient Matrix
L	Aerodynamic lift force [N]
Q	Dynamic pressure, $Q = \frac{1}{2}\rho V^2$ [PSF]
V_∞	Freestream wind speed [m s^{-1}]
W	Induced Downwash Matrix

Greek letters

α	Angle of attack [deg]
α_M	Main sail angle of attack [deg]
α_J	Jib sail angle of attack [deg]
Γ	Circulation [m^2/s]
ρ	Air density [kg m^{-3}]
ψ	Yaw angle [deg]
Ψ	Stream Function

1. INTRODUCTION

1.1. Background

Rovers have become a focus of innovation due to their ability to survey and explore remote, inaccessible, or uninhabited environments. Bio-inspired design approaches have played a significant role in shaping several rover concepts. One source of inspiration is the tumbleweed, which can traverse large distances by harnessing wind energy, making it an attractive model for a passive, energy-efficient rover. Several tumbleweed-inspired rover designs have been proposed in the literature. The rovers presented in [1], [2], [3], and [4] rely on drag-based propulsion for motion. Although these designs are highly energy efficient, drag-driven propulsion inherently limits controllability and prevents upwind travel. In contrast, the Land-Sailing Rover [5]

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employs lift-based propulsion, enabling upwind motion; however, this configuration is susceptible to failure under gusty wind conditions. A detailed comparison of these concepts and the associated trade-offs is presented in [6]. Motivated by this literature review, a new spherical rover concept has been proposed and is currently under development.

The Spherical Sailing Omnidirectional Rover (SSailOR) is a wind-powered autonomous rover, motivated by the growing need for persistent and sustainable robotic systems to operate in remote environments. The SSailOR is the first wind-powered spherical rover capable of net upwind motion. As shown in Fig. 1, this is a spherical terrestrial rover, inspired by the sailboat, with a centerboard equivalent and lifting sails. Its traction hoop serves as the centerboard, enabling movement on the given terrain. Long-duration mobility is enabled by wind propulsion via onboard sails. These sails are controllable for desired paths, including upwind motion. Potential applications for this concept are wide, including planetary exploration, arctic observation, and military surveillance. This rover is to have a design that generates lift with minimal energy consumption. This is a unique concept, which makes it complex and requires a well-designed rover.

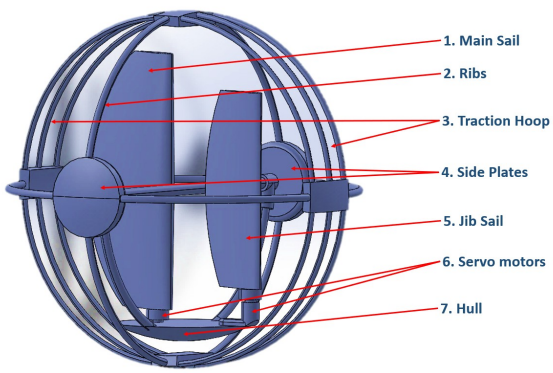

FIGURE 1: SSailOR schematic diagram

1.2. Numerical Methods and Applications

High-fidelity computational fluid dynamics (CFD) has become a standard tool for analyzing complex aerodynamic flows and structures due to its ability to handle viscous effects, flow separation, and unsteady aerodynamics. However, for an early-stage design and parametric study, CFD often presents significant limitations. Some of them include high computation cost associated with mesh generation, solver convergence, and post-processing, making design optimization expensive [7]. In contrast, low-fidelity aerodynamic models based on the potential flow theory offer substantial advantages in terms of computational efficiency while still capturing the required forces with decent accuracy. One such numerical technique is the Weissinger method, an extension of the classical lifting-line theory. It discretizes lifting surfaces into bound vortices with trailing vortex sheets and enforces a no-flow boundary condition at control points along the surface [8]. This approach enables the prediction of lift distribution, induced drag, and aerodynamic coupling effects at a fraction of the computational cost of CFD. For configurations where flow separation is limited and angles of attack remain moderate, i.e., within the stall range, the weissinger method has been shown to provide reliable accuracy, making it a go-to tool for parametric study or optimization.

The usefulness of the Weissinger method becomes particularly evident in multi-surface aerodynamic problems, such as tandem wings, canards, and multi-sail systems, where mutual interference plays a critical role in overall performance. By modeling the induced velocity field from each lifting surface, the method naturally captures aerodynamic interaction effects, including upwash, downwash, and lift amplification due to proximity and alignment. These capabilities make the Weissinger method an attractive choice for analyzing the coupled aerodynamics of SSailOR's main and jib sail configuration, where understanding relative sail placement, trim angles, and spacing is essential for maximizing lift, upwind motion, and control authority.

To verify and validate the Weissinger-based implementation, XFLR5 is employed as an independent numerical tool. XFLR5 is a widely used aerodynamic analysis software that integrates lifting-line theory (LLT), vortex lattice methods (VLM), and panel methods for the analysis of airfoils, wings, and sail-like lifting surfaces [9]. Its versatility has led to extensive adoption in sail aerodynamics, low-speed aircraft design, unmanned aerial vehicles, and wind-driven vehicle research. XFLR5 allows rapid evaluation of lift, drag, and moment coefficients across a wide range of operating conditions and provides an effective reference for validating custom aerodynamic models. While XFLR5 itself is not a high-fidelity CFD solver, its implementation of vortex-based methods has been validated against experimental data and higher-fidelity simulations in numerous studies, demonstrating its suitability for conceptual design and comparative analysis [10].

Overall, the combination of the Weissinger method and XFLR5 offers a powerful and computationally efficient framework for aerodynamic analysis and comparison. This approach enables rapid exploration of a system that is in an initial design phase and provides physical insight into aerodynamic behavior, which is difficult to achieve using CFD alone during early-stage development. As such, low-fidelity CFD methods play a critical role in guiding the design and optimization of wind-propelled robotic platforms like SSailOR before transitioning to higher-fidelity simulations or experimental testing.

1.3. Structure of the paper

This paper focuses on the aerodynamic interaction between the main sail and the jib sail, which dominates the lifting force, propulsion, and control authority of the SSailOR. The test matrix consisted of variations in the sail's yaw position ψ, main sail angle of attack α_M, jib sail angle of attackα_J, and wind speed V_∞ (in terms of dynamic pressure Q).

TABLE 1: Test Matrix

Parameter	Values Tested	Units
Rover heading (yaw)	20, 30, 40	degree
Main sail angle of attack	0 – 16	degree
Jib sail angle of attack	0 – 18	degree
Freestream wind speed	9.178, 11.241	m/s

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The test matrix is shown in the table 1. The main sail angle of attack was intentionally extended beyond its nominal stall angle to assess how aerodynamic interaction with the jib influences post-stall behavior and force production. Aerodynamic forces computed include lift L and drag D, where lift is defined as the force component perpendicular to the freestream wind direction and drag is defined as the force component aligned with the freestream. Both the Weissinger model and XFLR5 were evaluated independently using an identical test matrix to ensure consistency and enable direct comparison of predicted aerodynamic trends.

2. WEISSINGER METHOD

2.1. Methodology

In the Weissinger formulation, the lifting surface is discretized into a finite number of spanwise sections, each represented by a horseshoe vortex consisting of a bound vortex segment and trailing vortex legs extending downstream, shown in Fig. 2 [11]. To ensure consistency with thin airfoil theory, which predicts a lift-curve slope of 2π per radian (or 0.11 per degree) [12], the bound vortex of each section is placed along the quarter chord line, while the corresponding control point is located at the three-quarter chord position. At each control point, a zero normal-flow boundary condition is enforced, ensuring that the local flow remains tangent to the lifting surface.

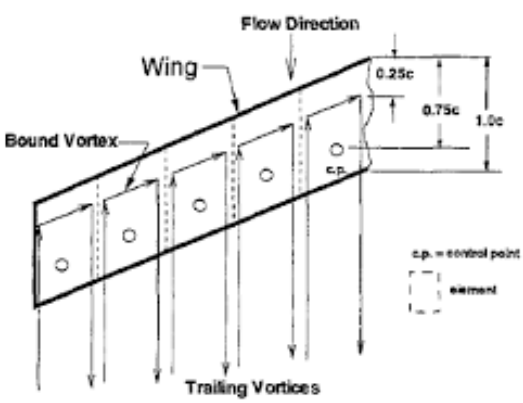

FIGURE 2: Weissinger Method Illustration

The normal flow velocity at a given control point consists of two contributions: the component of free stream velocity normal to the surface and the induced velocity generated by all horseshoe vortices in the system. Enforcing the zero normal-flow boundary condition requires the sum of these contributions to vanish at each control point.

$$V_{n,i} + (V_{\infty,n})_i = 0 \tag{1}$$

where $V_{n,i}$ is the induced normal velocity and $(V_{\infty,n})_i$ is the normal component of the freestream velocity at the ith control point. The freestream contribution depends on the local geometric orientation of the section and is expressed in terms of the geometric angle of attack, section twist, and zero-lift angle. The effective geometric angle of attack accounts for sweep through a trigonometric projection of the freestream velocity onto the section normal.

The induced velocity contribution in (1) is computed using the Biot-Savart law, accounting for the influence of all bound and trailing vortex segments on each control point[13]. To solve for the unknown circulation strengths, an aerodynamic influence coefficient (AIC) matrix is constructed. Each entry of the matrix represents the induced normal velocity at a given control point due to a unit-strength vortex at another section. The resulting linear system takes the form:

$$[F]\{\Gamma\} = \{V_n\} \tag{2}$$

where [F] is the N x N AIC matrix, where N is the number of horseshoe vortices, Γ is the vector of unknown circulations and (V_n) is the vector of freestream normal-velocity components. Figure 3 illustrates the concept of circulation about a lifting surface, where the bound vortex along the span induces downwash and generates lift. Solving equation (2) yields the circulation distribution along the span.

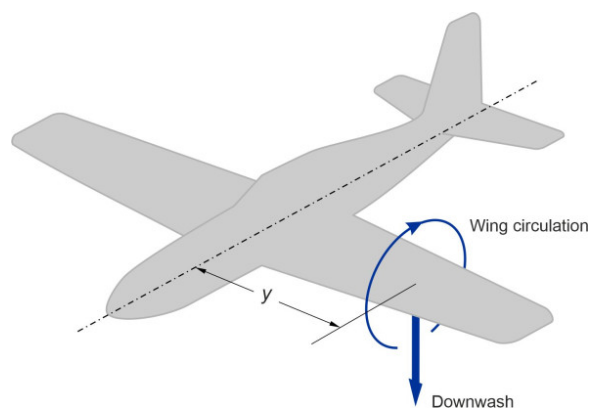

FIGURE 3: Circulation Around a Lifting Surface

Once the circulation distribution is obtained, sectional lift is computed using the Kutta- Joukowski theorem [14], assuming that induced velocity components in the freestream direction are small relative to the freestream magnitude. The elemental lift for each section is given by:

$$dL_i = \rho V_\infty \Gamma_i ds_i \tag{3}$$

where ds_i is the spanwise length of the section. Using equation (3), the sectional lift coefficient is then expressed as:

$$C_{l,i} = \frac{2}{c_i}\left(\frac{\Gamma_i}{V_\infty}\right) \tag{4}$$

And the total lift coefficient is obtained by integrating the circulation distribution across the span:

$$C_L = \frac{2}{S_{\text{ref}}} \sum_{i=1}^{N} \left(\frac{\Gamma_i}{V_\infty}\right) \cos(\theta_i)\, ds_i \tag{5}$$

where θ_i is the local dihedral angle. The induced drag is computed from the product of circulation and downwash at each

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section. The induced drag coefficient is expressed as:

$$C_{D_i} = \frac{2}{S_{\text{ref}}} \sum_{i=1}^{N} \left(\frac{w_i}{V_\infty}\right)\left(\frac{\Gamma_i}{V_\infty}\right) ds_i \tag{6}$$

where W_i represents the normal downwash velocity at the section and is calculated using:

$$[W] = [F_m]\{\Gamma\} \tag{7}$$

where $[F_m]$ is the N x N AIC matrix calculated at the mid-point instead of the control point.

In this work, the Weissinger method provides an efficient framework for modeling the aerodynamic interaction between multiple lifting surfaces[15].

2.2. Application of Weissinger Method to the Dual-Sail Configuration

A schematic of the discretized sail configuration, including bound vortices, control points, and trailing vortices, is shown in Fig 4. To apply the Weissinger method to the SSailOR's dual-sail system, the formulation was implemented in a modular manner, divided into 4 primary stages: 1. Initialization, 2. Geometric data generation, 3. AIC matrix assembly, 4. Force computation.

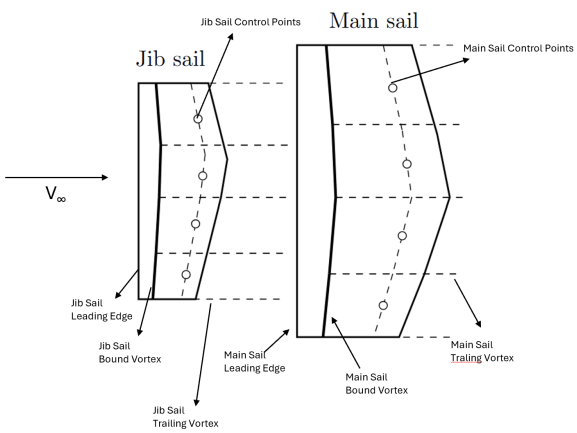

FIGURE 4: Weissinger Model Diagram For Sails

2.2.1. Initialization The first stage of the implementation consists of reading a structured input file that defines the global aerodynamic parameters and the geometric description of each sail. This file-based approach allows rapid modification of sail geometry and operating conditions without altering the solver logic, making it particularly suitable for parametric studies and design optimization.

At the global level, the input file specifies reference quantities such as the reference area S_{ref}, reference span B_{ref}, reference chord C_{ref}, and symmetry options. The lifting surfaces are then defined as a collection of patches, where each patch represents a spanwise segment of either the main sail or jib sail. For each patch, the number of spanwise panels and the geometric endpoints are specified. These endpoints define the local chord length, zero-lift angle, and twist, enabling the modeling of nonuniform sail geometry.

Finally, the operating condition is defined through the global angle of attack. This structure allows the same framework to be used for multiple yaw orientations by rotating the sail geometry relative to the freestream direction. The reading inputs table used for this case is shown in Table 2. The definitions of the parameters used in Table 2 are listed below and are shown schematically in Fig. 5 for clarity.

a = distance from jib sail LE to the axis of rotation of jib sail
b= distance from the jib sail axis of rotation to the main sail axis of rotation
c = distance from main sail LE to the axis of rotation of main sail
$\boldsymbol{\theta}_j = \psi - \alpha_j$

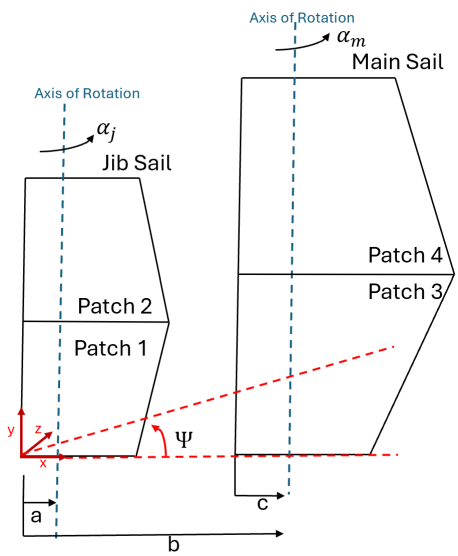

FIGURE 5: Geometry of Sails for Weissinger (Isometric Image)

2.2.2. Geometric Data Generation Once the input data are read, the geometry of each sail patch is processed to generate the Weissinger vortex system. Each patch is discretized into a prescribed number of spanwise panels. For each panel, the following parameters are computed:

- Quarter-chord points define the location of the bound vortices
- Three-quarter-chord points, which serve as control points
- Panel spanwise length and orientation

The bound vortex for each panel is placed along the quarter-chord line, extending between the panel's spanwise endpoints. From each bound vortex endpoint, a trailing vortex leg is extended downstream in the direction of the freestream velocity. This formulation naturally allows the induced velocity field from the jib sail to influence the main sail, and vice versa, enabling accurate modeling of coupled aerodynamics. All parameters for each panel are computed in a global reference frame, ensuring that changes in yaw angle are consistently reflected in the induced velocity calculations.

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TABLE 2: Input Parameters and Patch Geometry for Weissinger-Based Dual-Sail Model

Parameter	Value	Units
Symmetry	None	–
Number of patches, n_{patch}	4	–
Panels per patch, n_{pan}	50 (each patch)	–
Patch 1 (Jib) quarter-chord endpoints	$(x, y, z, c) = (0, 0, 0, 73)$ to $(0, 110, 0, 100)$	mm
Patch 2 (Jib) quarter-chord endpoints	$(x, y, z, c) = (0, 110, 0, 100)$ to $(0, 220, 0, 73)$	mm
Patch 3 (Main) quarter-chord endpoints	$(x, y, z, c) = (a + b\cos\theta_j,\ 0,\ b\sin\theta_j,\ 82)$ to $(a + b\cos\theta_j,\ 143,\ b\sin\theta_j,\ 160)$	mm
Patch 4 (Main) quarter-chord endpoints	$(x, y, z, c) = (a + b\cos\theta_j,\ 143,\ b\sin\theta_j,\ 160)$ to $(a + b\cos\theta_j,\ 286,\ b\sin\theta_j,\ 82)$	mm

2.2.3. AIC Matrix Assembly The third stage involves constructing the aerodynamic influence coefficient (AIC) matrix. Each element of the AIC matrix represents the induced normal velocity at a given control point due to a unit-strength horseshoe vortex located on another panel. The induced velocity contributions from the bound vortex segment and both trailing vortex legs are computed using the Biot–Savart law. These induced velocities are projected onto the local surface normal at each control point. The freestream normal velocity component is then added to enforce the zero normal-flow boundary condition. Assembling these contributions for all control points and vortex elements yields a square linear system of the form expressed in equation 2. Solving this system provides the circulation distribution across both sails simultaneously, inherently capturing aerodynamic coupling effects.

2.2.4. Force Computation In the final stage, the circulation distribution obtained from the linear system is used to compute aerodynamic forces. Sectional lift is calculated using the Kutta–Joukowski theorem (equation 3), assuming negligible induced velocity components in the freestream direction. The lift contributions from all panels are then integrated to obtain the total lift coefficient (equation 5). Induced drag is computed from the product of circulation and local downwash at each panel, allowing assessment of drag penalties associated with sail interaction and trim changes (equation 6). The resulting lift and drag forces are resolved relative to the freestream direction, enabling direct comparison with XFLR5 output and future experimental measurements.

2.3. Weissinger Method Results

The results from the Weissinger method are shown in Figure 6 and Figure 7. It predicts a consistent increase in lift with increasing main sail angle of attack across all yaw angles, while the induced drag remains comparatively small and grows gradually. Higher yaw angles result in increased lift for a given sail trim, driven by greater effective flow incidence and enhanced coupled aerodynamics effects.

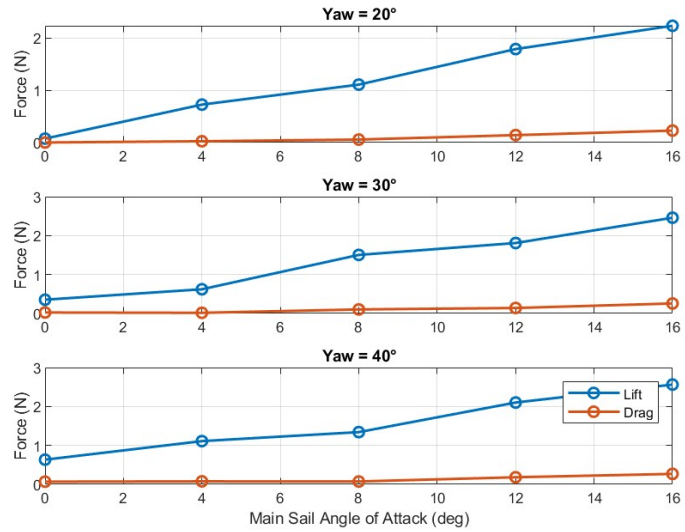

FIGURE 6: Weissinger At 1 PSF

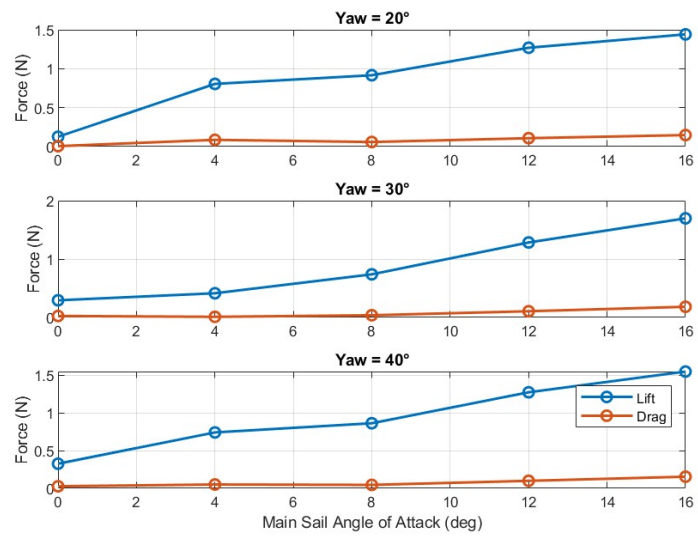

FIGURE 7: Weissinger At 1.5 PSF

3. ANALYSIS USING XFLR5

3.1. Methodology

XFLR5 is a program that puts together other software, specifically XFOIL and PROFOIL, for the design and simulation of airfoils, wings, and aircraft. This program is supported by mathematical methods, including the Vortex Lattice Method, Lifting Line Theory, and the 3D Panel Method, to model the macroscopic effects and to analyze behavior of aircraft [16], [17], [18], [19]. XFLR5 solves for aerodynamic performance at low Reynolds numbers, capable of gathering lift, drag, pressure, and pitching moment. XFLR5 is a useful tool for analyzing 2D airfoils, finite-wings, and airplanes.

The sails simulated are in inviscid flow. Inviscid flow disregards the friction effects in the fluid and at the boundary. In XFLR5, the airfoil is first made in XFOIL. XFOIL analyzes in-

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viscid fluid in the following procedure, further explained in [17], [20]. The aerodynamic effect of an airfoil is determined by solving along the airfoil using "nodes." A 2D streamline function for a given geometry of a vortex sheet on the airfoil with a source sheet on the wake and surface is expressed.

$$\Psi(x, y) = u_\infty - v_\infty + \frac{1}{2\pi}\int \gamma(s)\ \ln r(s; x, y)\, ds + \frac{1}{2\pi}\int \sigma(s)\, \theta(s; x, y)\, ds$$

For a constant value of Ψ_0 across several nodes, this stream function can be solved into a linear system.

$$\sum_{j=1}^{N} a_{i,j}\, \gamma_j - \Psi_0 = -u_\infty\, y_i + v_\infty\, x_i - \sum_{j=1}^{N} b_{i,j}\, \sigma_{0,j}, \qquad 1 \le i \le N$$

The additional coefficients $a_{i,j}$ and $b_{i,j}$ can be determined from the streamline function when the panel nodes x_i and y_j and the wake nodes are known. Applying the Kutta condition simplifies the expression as shown below.

$$\gamma_1 + \gamma_N = 0$$

Since the flow inside the airfoil is stagnant, the surface vorticity and velocity are equal. Using Bernoulli's equation, the pressure coefficient can be expressed in terms of the surface vorticity. From here, the other elements of lift and drag are straightforward to solve.

$$C_p = \frac{\gamma}{q_\infty}$$

3.2. Application of XFLR5 to the dual-sail configuration

To verify the Weissinger method results, an independent aerodynamic analysis was performed using XFLR5. Since XFLR5 is formulated around aircraft lifting surfaces, the SSailOR system was mapped onto an equivalent configuration by modeling the jib sail as the main wing and the main sail as an elevator. This approach enables the use of XFLR5's built-in vortex lattice and lifting-line solvers while preserving the relative positioning and aerodynamic interaction between the two sails.

For all yaw cases, the jib sail (main wing) was placed at the reference origin, with its leading edge located at x = 0 and z = 0. The main sail (elevator) was positioned downstream and offset relative to the jib sail. The geometric offset between the leading edges of the jib and main sails was maintained at a fixed distance of 0.12 m, corresponding to the physical separation in the SSailOR configuration.

3.2.1. 20 Degree Yaw For the 20-degree yaw case, the jib sail (main wing) was fixed at the reference origin with its leading edge located at x=0 and z=0. The corresponding main sail (elevator) location was computed using simple trigonometric relationships:

$$z = 0.12 \tan(20°) \tag{8}$$

$$x = \frac{0.12}{cos(20°)} \tag{9}$$

This configuration preserves the correct projected spacing between the sails when exposed to a 20° yawed freestream, as illustrated in Fig. 8

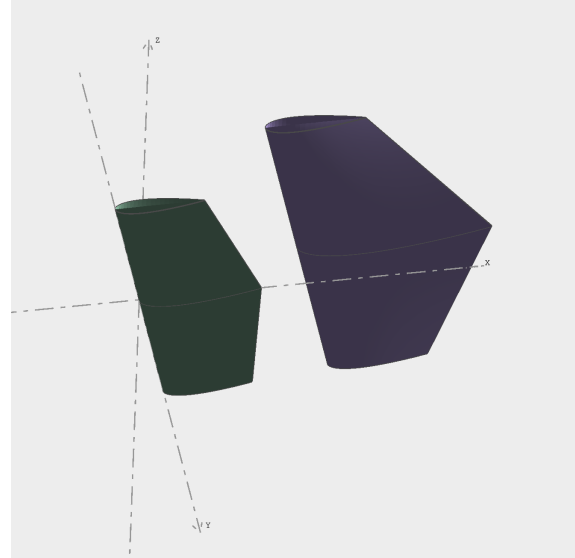

FIGURE 8: SSailOR at 20 Degree Yaw in XFLR5

3.2.2. 30 degree yaw For the 30-degree yaw case, the jib sail remained fixed at x=0, z=0, while the main sail position was adjusted to reflect the increased yaw angle. Using the same geometric separation of 0.12 m, the elevator offsets were determined as:

$$z = 0.12 \tan(30°) \tag{10}$$

$$x = \frac{0.12}{cos(30°)} \tag{11}$$

As shown in Fig. 9, this results in a larger vertical offset and downstream placement of the main sail, increasing the effective interaction between 2 surfaces under 30° yawed freestream

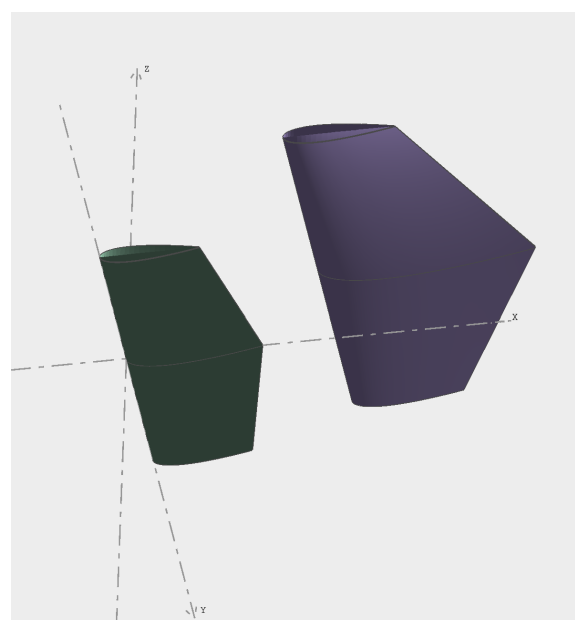

FIGURE 9: SSailOR at 30 Degree Yaw in XFLR5

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3.2.3. 40 dergree yaw For the 40-degree yaw case, the same modeling approach was applied, with jib sail fixed at the origin and the main sail positioned using:

$$z = 0.12\tan(40^\circ) \tag{12}$$

$$x = \frac{0.12}{cos(40^\circ)} \tag{13}$$

This configuration produces the largest geometric offset between the sails, as illustrated in Fig.10, and represents the most aggressive yaw condition examined in this study. Despite the increased separation, the mapping preserves geometric consistency with the Weissinger-based model and enables meaningful aerodynamic comparison.

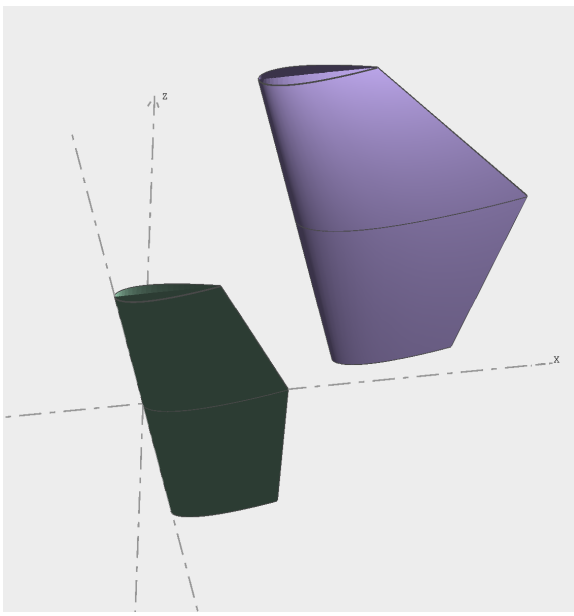

FIGURE 10: SSailOR at 40 Degree Yaw in XFLR5

3.3. Results of XFLR5

The aerodynamic forces computed by XFLR5 for the dual-sail configuration are shown in Fig. 11 and Fig. 12 for yaw angles of 20°, 30°, and 40°. Across all yaw cases, lift increases with the main sail angle of attack, while drag remains significantly smaller and grows gradually. Higher yaw angles increase lift for a given sail trim, consistent with the increased effective flow incidence and enhanced interaction between the jib (main wing) and the main sail (elevator). The lift trends remain smooth and largely linear over the examined angle-of-attack range, indicating stable numerical behavior of the XFLR5 vortex-based solver.

4. COMPARISON

4.1. Plots and Observation

Fig. 13, Fig. 14, Fig. 15, Fig. 16 compare the lift and drag predictions obtained from the weissinger method and XFLR5 across yaw angles of 20°, 30°, and 40°. Overall, the two methods show strong agreement in both lift and drag trends over the examined range of main sail angles of attack. Lift predictions from the

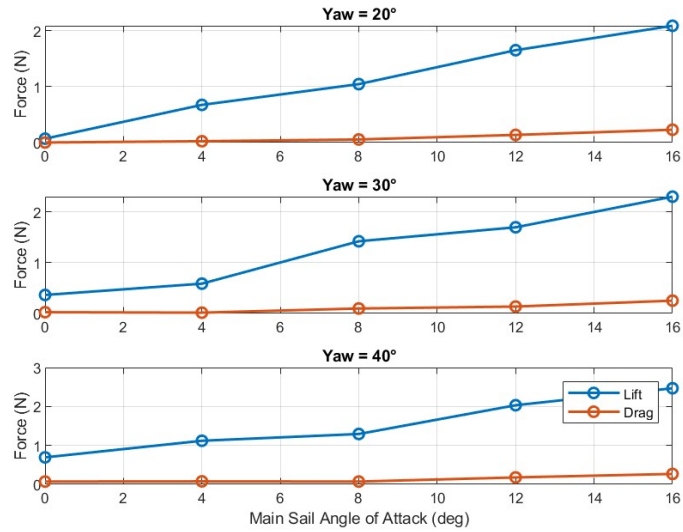

FIGURE 11: XFLR5 At 1 PSF

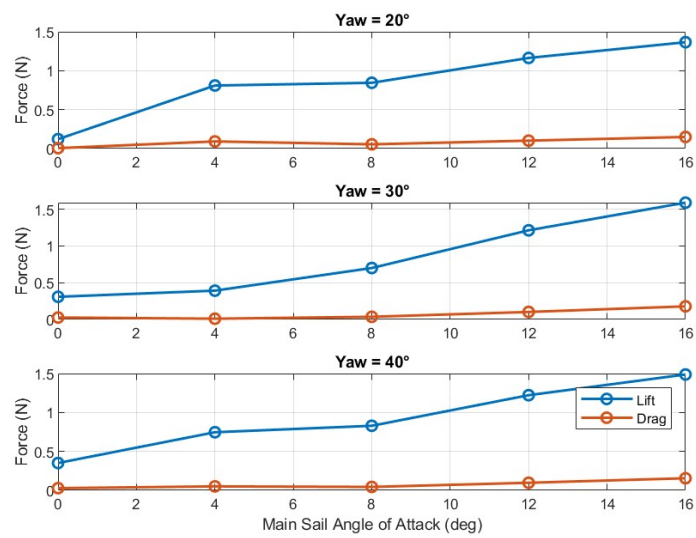

FIGURE 12: XFLR5 At 1.5 PSF

Weissinger model closely match those from XFLR5 for all yaw cases, with minor deviations appearing at higher angles of attack, where increased induced effects and modeling assumptions begin to influence the results.

Drag predictions also show good agreement, with both methods capturing the gradual increase in drag with angle of attack and yaw angle. Small discrepancies are observed at higher yaw angles and larger sail trims, which can be attributed to differences in how induced drag and multi-surface interaction are modeled in the two approaches. Despite these differences, the Weissinger method reliably reproduces the overall aerodynamic behavior predicted by XFLR5, validating its use as a fast and effective reduced-order tool for parametric analysis and early-stage design of the dual-sail configuration.

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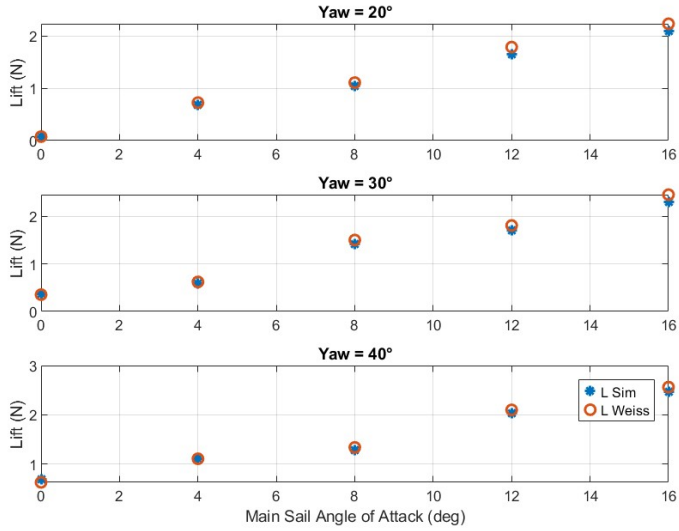

FIGURE 13: Lift Comparison @ 1 PSF

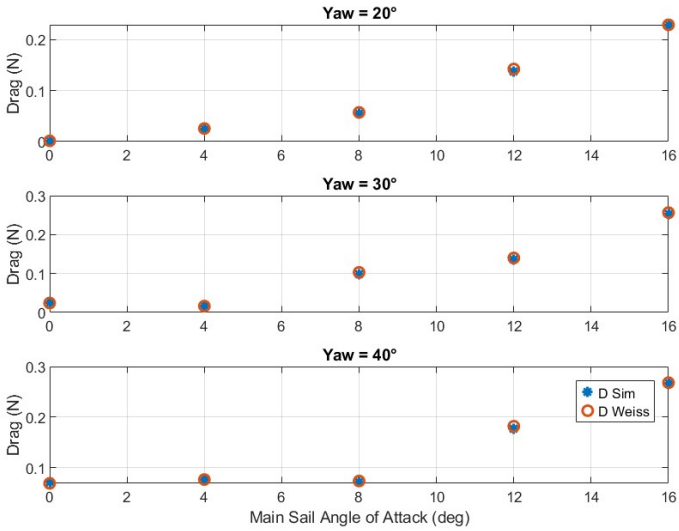

FIGURE 14: Drag Comparison @ 1 PSF

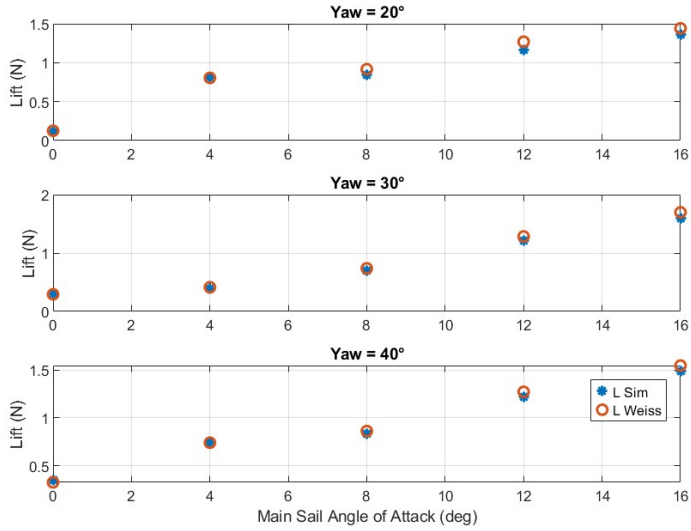

FIGURE 15: Lift Comparison @ 1.5 PSF

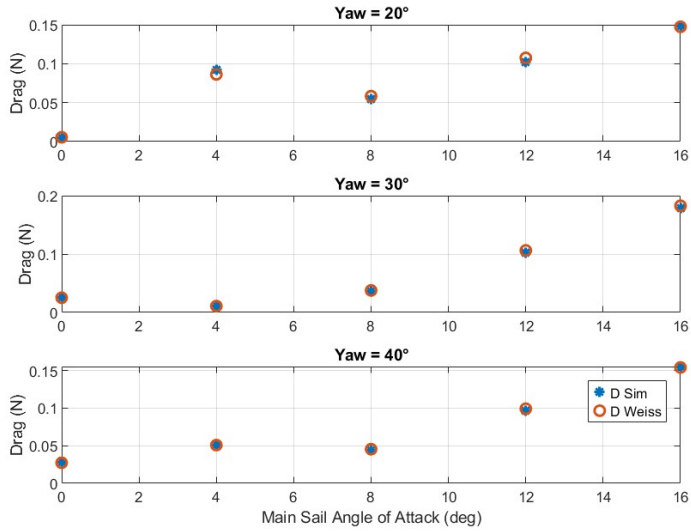

FIGURE 16: Drag Comparison @ 1.5 PSF

4.2. Discussion

While the weissinger method and XFLR5 methods show consistent aerodynamic trends, the remaining differences highlight the limitations of low-fidelity models for interacting lifting surfaces. The weissinger method assumes inviscid flow and simplified wake behavior, which limits its ability to fully capture nonlinear effects that become more dominant at higher yaw and sail angles. These assumptions primarily influence induced drag prediction and the distribution of circulation across the sails.

Additional discrepancies arise from the geometric abstraction required in XFLR5, where the dual-sail system is mapped onto a wing-elevator configuration. Although this approach preserves relative positioning and interaction effects, it introduces simplifications in wake alignment and surface coupling that affect force computation. Despite these limitations, the observed differences remain within expected bounds for low-fidelity models, making weissinger a useful tool for rapid parametric studies and early-stage design, while motivating higher-fidelity simulations and experimental validation in future work.

5. CONCLUSION

This work presented an aerodynamic analysis of a dual-sail configuration for the SSailOR using the Weissinger method and validated using XFLR5. The results demonstrate that the Weissinger model captures the dominant aerodynamic trends associated with sail trim and yaw, showing strong agreement with XFLR5 across a broad operating range. These findings confirm the suitability of the Weissinger method as a fast and effective tool for early-stage design and parametric exploration of wind-propelled robotic platforms. Future work will focus on higher-fidelity CFD simulations and experimental validation to further refine the aerodynamic model and quantify effects beyond the scope of reduced-order methods.

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